

A closed generating cone whose extension norm does not preserve the ambient topology

A full negative answer to the natural question in arXiv:2508.17052

Autonomous proof audit — lane 9

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Candidate full result. The two norm topologies in the source need not agree. There is an explicit proper, norm-closed, generating cone F in the incomplete normed space c_{00} for which vectors tending to zero in the ambient Euclidean norm stay uniformly away from zero in the cone-extension norm. The obstruction is exactly failure of a uniform bounded-decomposition estimate.

1 Source question and scope

Let F be a real linear cone and let $V = \text{span}(F)$ carry a norm $\mathbf{n}$. Proposition 2.20 of Kharitonov–Ohanyan defines

$$\tilde{\mathbf{n}}(x) = \inf\{\mathbf{n}(u) + \mathbf{n}(v) : u, v \in F, x = u - v\}. \quad (1)$$

The paper proves that this is a norm and that it agrees with $\mathbf{n}$ on F . Immediately after Lemma 2.22 it asks whether the norm topologies of $\mathbf{n}$ and $\tilde{\mathbf{n}}$ agree. Lemma 3.32 later proves that they do when F has nonempty $\mathbf{n}$ -interior, but does not settle the general question.

It remains to prove that $\text{quot} \cong \text{quot}$. We show that the identity map $\text{id}_{\text{span}(F)} : \text{quot} \rightarrow \text{quot}$ is continuous. By the universal property of quotient maps the map $\text{id}_{\text{span}(F)}$ is continuous if and only if the map $\text{id}_{\text{span}(F)} \circ q = q$ is continuous, where $q : F \times F \rightarrow V_{\tau_{\mathbf{n}}}$ is the quotient map $q(x, y) = \iota(x) - \iota(y)$. Note that $F \times F$ is given the product topology of $\tau_{\mathbf{n}}$ with itself. It suffices to show that $\iota : (F, \mathbf{n}) \rightarrow V_{\tau_{\mathbf{n}}}$ is continuous which follows from the fact that

$$\tilde{\mathbf{n}}(x) \leq \mathbf{n}(x) + \mathbf{n}(0) = \mathbf{n}(x)$$

for all $x \in F$. □

Lemma 2.22. *In the setting of Theorem 2.21, $\mathbf{n}$ agrees with $\tilde{\mathbf{n}}$ on F .*

Proof. Let $x \in F$. As shown in Theorem 2.21, $\tilde{\mathbf{n}}(x) \leq \mathbf{n}(x) + \mathbf{n}(0) = \mathbf{n}(x)$. On the other hand, it is always true that $\mathbf{n}(x) \leq \tilde{\mathbf{n}}(x)$ for all $x \in \text{span}(F)$. □

We left open the natural question whether the norm topologies of $\mathbf{n}$ and $\tilde{\mathbf{n}}$ agree. This is addressed further below in Lemma 3.32.

3 Positively polarizable hyperbolic norms

Figure 1: The natural question in arXiv:2508.17052, printed page 12.

The answer in general is no, even if the cone is proper, closed, and generating. Completeness is the missing hypothesis in the example below.

2 The bounded-decomposition criterion

The elementary criterion behind the question is useful for locating the obstruction.

Proposition 1. *For a generating cone F in a normed space $(V, \mathbf{n})$, the norms $\mathbf{n}$ and $\tilde{\mathbf{n}}$ induce the same topology if and only if there is a constant $C < \infty$ such that*

$$\tilde{\mathbf{n}}(x) \leq C\mathbf{n}(x) \quad (x \in V).$$

Equivalently, for some $C' < \infty$, every $x \in V$ has a decomposition $x = u - v$, $u, v \in F$, satisfying

$$\mathbf{n}(u) + \mathbf{n}(v) \leq C'\mathbf{n}(x).$$

Proof. The triangle inequality gives $\mathbf{n}(x) \leq \mathbf{n}(u) + \mathbf{n}(v)$ for every admissible decomposition, hence $\mathbf{n} \leq \tilde{\mathbf{n}}$. Thus equality of the two norm topologies is equivalent to continuity of the identity from $(V, \mathbf{n})$ to $(V, \tilde{\mathbf{n}})$, which is equivalent to the displayed uniform inequality. The decomposition formulation follows from the definition of the infimum: the uniform inequality gives actual decompositions with any fixed larger constant, and the converse is immediate. $\square$

3 An explicit counterexample

Let $V = c_{00}(\mathbb{N}; \mathbb{R})$, the real finitely supported sequences, with the Euclidean norm

$$\mathbf{n}(x) = \left(\sum_{k \geq 1} |x_k|^2 \right)^{1/2}.$$

Write (e_k) for the standard unit vectors and, for $j \geq 1$, set

$$f_{2j-1} = e_{2j-1}, \quad f_{2j} = e_{2j-1} + \frac{1}{j}e_{2j}.$$

The sequence (f_k) is an algebraic basis of V . Let

$$F = \left\{ \sum_k a_k f_k : a_k \geq 0 \text{ and only finitely many } a_k \neq 0 \right\}.$$

Theorem 1. *The set F is a proper, $\mathbf{n}$ -closed, generating real linear cone in V , and $\text{Int}_{\mathbf{n}}(F) = \emptyset$. Nevertheless, if $\tilde{\mathbf{n}}$ is defined by (1), then $\mathbf{n}$ and $\tilde{\mathbf{n}}$ do not induce the same topology. Indeed, for*

$$x_j = f_{2j} - f_{2j-1} = \frac{1}{j}e_{2j}$$

one has

$$\mathbf{n}(x_j) = \frac{1}{j} \longrightarrow 0, \quad \tilde{\mathbf{n}}(x_j) \geq 2.$$

Proof. Because (f_k) is an algebraic basis, nonnegative coefficient expansions are unique. It follows at once that $F \cap (-F) = \{0\}$ and $F - F = V$, so F is proper and generating. In standard coordinates,

$$F = \{x \in c_{00} : x_{2j} \geq 0 \text{ and } x_{2j-1} \geq jx_{2j} \text{ for every } j\}.$$

Every coordinate functional is continuous for $\mathbf{n}$, so this intersection of closed half-spaces is closed in V . It has empty interior: given $x \in F$, choose j outside its finite support. Then $x + \delta e_{2j} \notin F$ for every $\delta > 0$, although its distance from x is δ .

It remains to estimate the extension norm. Fix j and suppose $x_j = u - v$ with $u, v \in F$. Write

$$u = \sum_k \alpha_k f_k, \quad v = \sum_k \beta_k f_k, \quad \alpha_k, \beta_k \geq 0.$$

Uniqueness of the f -coordinates and $x_j = f_{2j} - f_{2j-1}$ give

$$\alpha_{2j} - \beta_{2j} = 1, \quad \alpha_{2j-1} - \beta_{2j-1} = -1.$$

Consequently $\alpha_{2j} \geq 1$ and $\beta_{2j-1} \geq 1$. The $(2j-1)$ st standard coordinates of u and v are respectively

$$u_{2j-1} = \alpha_{2j-1} + \alpha_{2j} \geq 1, \quad v_{2j-1} = \beta_{2j-1} + \beta_{2j} \geq 1.$$

Therefore $\mathbf{n}(u) + \mathbf{n}(v) \geq 2$ for every admissible decomposition. Taking the infimum proves $\tilde{\mathbf{n}}(x_j) \geq 2$, whereas $\mathbf{n}(x_j) = 1/j$. Hence x_j converges to zero in one norm topology and not in the other. $\square$

For completeness, the displayed lower bound is nearly sharp: the defining decomposition gives

$$\tilde{\mathbf{n}}(x_j) \leq \mathbf{n}(f_{2j}) + \mathbf{n}(f_{2j-1}) = \sqrt{1 + j^{-2}} + 1.$$

Thus $\tilde{\mathbf{n}}(x_j) \rightarrow 2$ while $\mathbf{n}(x_j) \rightarrow 0$.

4 Why the hypotheses are sharp

The source's nonempty-interior argument produces the bounded decomposition in Proposition 1. The counterexample shows that empty interior alone permits failure, even with a closed cone. It also pinpoints why the ambient space was chosen to be incomplete: Andô's bounded decomposition theorem states that a closed generating cone in a Banach space admits a constant ρ such that every x can be written $x = u - v$ with u, v in the cone and $\mathbf{n}(u), \mathbf{n}(v) \leq \rho \mathbf{n}(x)$ [2, Lemma 1]. Proposition 1 then gives $\tilde{\mathbf{n}} \leq 2\rho \mathbf{n}$. Hence:

- in the general normed-space setting of the source, the answer is negative by Theorem 1;
- if the ambient normed space is complete and F is closed and generating, the answer is affirmative, with no interior assumption.

5 Verification and novelty status

The construction was audited directly in the source's conventions: F is a cancellative cone, its canonical span is identified with V , and (1) is exactly the source's extension norm. The lower bound uses only uniqueness of finite basis expansions and two standard-coordinate projections. No limiting point outside c_{00} is invoked in the proof of closedness, which is relative to the stated normed space.

Bounded exact-phrase, title, generating-cone, bounded-decomposition, and Andô-theorem searches found the source question and the classical positive theory, but no later paper explicitly answering this particular question or this block-basis counterexample. The underlying phenomenon is standard in ordered normed-space theory, so novelty is claimed only for the explicit answer and its connection to arXiv:2508.17052. Expert or author review is recommended before dissemination.

Lemma 3.32. *If $\text{Int}(F) \neq \emptyset$ then $\tilde{\mathbf{n}}$ and $\mathbf{n}$ are equivalent as norms on $\text{span}(F)$.*

Proof. For any $u, v \in F$ with $x = u - v$, we have $\mathbf{n}(x) \leq \mathbf{n}(u) + \mathbf{n}(v)$. Taking the infimum over all such u, v , we get $\mathbf{n}(x) \leq \tilde{\mathbf{n}}(x)$. Now suppose that there exists some $s \in \text{Int}(F)$. That is, there exists a $\delta > 0$ such that $v \in F$ for all $v \in \text{span}(F)$ with $\mathbf{n}(s - v) < \delta$. We can now show that $\tilde{\mathbf{n}}(x) \leq K\mathbf{n}(x)$ for all $x \in \text{span}(F)$ and some $K > 0$. If $\tilde{\mathbf{n}}(x) = 0$ then $x = 0$ and $\tilde{\mathbf{n}}(x) = 0$ so we are done. Otherwise, let $\varepsilon = \frac{\delta}{2\mathbf{n}(x)}$, $y_+ = s + \varepsilon x$, $y_- = s - \varepsilon x$. Since

$$\mathbf{n}(s - y_+) = \mathbf{n}(s - y_-) = \mathbf{n}(\varepsilon x) = \frac{\delta}{2} < \delta,$$

it follows that $y_+, y_- \in F$. Note that $y_+ - y_- = 2\varepsilon x$, so

$$x = \frac{y_+ - y_-}{2\varepsilon}.$$

Let $u = \frac{y_+}{2\varepsilon}$ and $v = \frac{y_-}{2\varepsilon}$. This gives us $x = u - v$ with $u, v \in F$. By definition of $\mathbf{n}$,

$$\tilde{\mathbf{n}}(x) \leq \mathbf{n}(u) + \mathbf{n}(v) = \frac{\mathbf{n}(y_+) + \mathbf{n}(y_-)}{2\varepsilon}.$$

By the triangle inequality,

$$\mathbf{n}(y_+) + \mathbf{n}(y_-) = \mathbf{n}(s + \varepsilon x) + \mathbf{n}(s - \varepsilon x) \leq 2\mathbf{n}(s) + 2\varepsilon\mathbf{n}(x).$$

Thus,

$$\tilde{\mathbf{n}}(x) \leq \frac{\mathbf{n}(s) + \varepsilon\mathbf{n}(x)}{\varepsilon} = \frac{2\mathbf{n}(x)\mathbf{n}(s)}{\delta} + \mathbf{n}(x) = \left(\frac{2\mathbf{n}(s)}{\delta} + 1\right)\mathbf{n}(x).$$

Taking $K = \frac{2\mathbf{n}(s)}{\delta} + 1$ completes the proof. $\square$

Figure 2: The source's positive result under nonempty interior, printed page 22.

References

- [1] E. Kharitonov and A. Ohanyan, *Topological cones and positively polarizable hyperbolic norms*, arXiv:2508.17052 (2025), Proposition 2.20, Lemma 2.22, and Lemma 3.32.
- [2] T. Andô, *On fundamental properties of a Banach space with a cone*, Pacific J. Math. **12** (1962), 1163–1169, doi:10.2140/pjm.1962.12.1163.